

Automobiles with Vision: What We Can Look Forward To

Imagine how difficult it would be to live without sight—the most amazing ability we humans possess. But there are machines that can see, for us. This concept, dreamt of for years, is now ready and deployed. System designers have used embedded vision technology to incorporate exciting vision capabilities in automobiles. Let us dive in to find out more

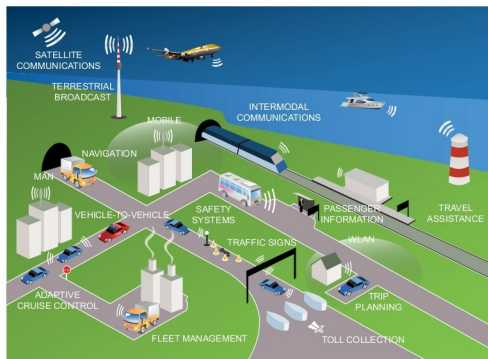


Fig. 1: Intelligent transportation system—a typical function of an ADAS

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Embodied vision technology is based on visual input by using embedded systems. It is basically a combination of two things—embedded systems and computer vision technology.

Once limited to minor use, the field of embodied vision is growing and has now spread to more and more applications. New high-volume markets for this technology include automotive driver assistance, home surveillance and gaming systems. Embedded vision technology within an automobile can serve to enhance one's driving experience immensely.

Technologies that keep drivers

alert, safe and accident-free are known as advanced driver assistance systems, or ADAS (a typical function of which can be seen in Fig. 1). Automobiles with such technologies are intelligent vehicles on which sensors, cameras and control systems are integrated to assist in the task of driving. They increase car and road safety. The aim is to combine sensors, cameras and algorithms to understand the vehicle environment so that the driver can receive assistance or be warned of potential hazards. ADAS are usually safety focussed, but are mostly marketed as a convenience. Cameras are the most used sensors in these systems—which can also be called machine-vision systems or intelligent systems.

Cameras and video analytics are used for various applications in automobiles. Advanced vehicles equipped with camera vision use one or multiple cameras to detect and recognise vehicles, pedestrians, traffic signs and so on, around the vehicle, apart from the state of the driver and passengers, with the help of image recognition technologies.

Video content analysis (VCA) is the process to automatically analyse video to detect and determine temporal events not based on a single image. It uses software algorithms for the analysis of CCTV images to detect alarming conditions, such as an intruder moving into a restricted area. The algorithms can be implemented as software on general-purpose systems, or as hardware in specialised video processing units. The major benefit of this technology is the potential for automating the sometimes laborious task of monitoring.

In ADAS, the combination of these technologies makes the driver aware (by using a visual or audio alert) of an object or pedestrian that is approaching really fast. In some cases, the vehicle itself takes some action, such as applying the brakes. The most useful video analytics software can detect objects of a specific size reliably—typically people—whilst ignoring irrelevant objects, and generate an alarm only when specifically set conditions are met, which are configured in the software by the user.

“ADAS represent a major emerging market trend of using video devices inside ‘current and future automobiles;’